

⇒ Find the product of

$$\left(\frac{3b}{2} + \frac{2}{3b} \right) \left(\frac{9b^2}{4} + \frac{4}{9b^2} - 1 \right)$$

$$(a^3 + b^3) = (a+b)(a^2 - ab + b^2)$$

$$\left(\frac{3b}{2} + \frac{2}{3b} \right) \left(\frac{3b}{2} \right)^2 - \left(\frac{3b}{2} \right) \left(\frac{2}{3b} \right) + \left(\frac{2}{3b} \right)^2$$

$$\left(\frac{3b}{2} + \frac{2}{3b} \right) \left(\frac{9b^2}{4} \right) - 1 + \left(\frac{4}{9b^2} \right)$$

$$\left(\frac{3b+2}{2 \cdot 3b} \right) \left(\frac{9b^2}{4} + \frac{4}{9b^2} - 1 \right) = \frac{\left(\frac{3b}{2} \right)^3 + \left(\frac{2}{3b} \right)^3}{(a)^3 + (b)^3}$$

$$\frac{3b+2}{2 \cdot 3b} \left(\frac{9b^2}{4} + \frac{4}{9b^2} - 1 \right) = \frac{27b^3}{8} + \frac{8}{27b^3}$$

Ans: